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Artificial Intelligence and Machine Learning

(6CS012)

Task 1 Report

Question and Answer

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Group: L6CG18

Supervisor: Mr Shiv Kumar Yadav

Module Leader: Simon Giri

1. Long Question (Answer any 1 — 5 Marks)

Option A: Unsupervised Learning for a Payment Platform

You are a Data Scientist at a leading digital payment platform such as eSewa, working on large-scale user transaction data.

• Identify two high-impact business problems where unsupervised learning can provide value (e.g customer segmentation, fraud detection, behavioral profiling).

• For each problem:

– Clearly define the problem in a real-world context.

– Propose a suitable unsupervised learning approach (e.g., clustering, anomaly detection, dimensionality reduction) and recommend specific algorithms (e.g., K-Means, DBSCAN, Autoencoders).

– Discuss how the model outputs can be integrated into real-time or batch systems to support business decisions.

– Briefly mention one limitation or challenge of your proposed approach.

• Assume limited labeled data and large-scale transactions.

Ans-

Problem 1: Customer Segmentation for Personalized Marketing

Problem Definition: eSewa processes millions of transactions daily across diverse user demographics — students, merchants, freelancers, and rural users. A one-size-fits-all marketing approach wastes resources and reduces engagement. The challenge is to group users into meaningful behavioral segments without predefined labels, based on transaction frequency, average transaction value, merchant categories, time-of-day patterns, and feature usage (bill payments, remittances, QR payments).

Proposed Approach: A hybrid clustering strategy works best here. **K-Means** provides fast, scalable initial segmentation into broad groups (e.g., high-value frequent users, occasional low-value users, dormant users). For users with irregular or overlapping patterns, **DBSCAN** handles non-spherical clusters and naturally identifies outliers. Before clustering, **PCA (Principal Component Analysis)** reduces the high-dimensional transaction feature space, removing noise and improving cluster quality.

Business Integration: Model outputs can run as a **weekly batch pipeline** on transaction data warehouses. Segment labels are pushed to CRM and campaign management tools, enabling targeted promotions — cashback offers for dormant users, loyalty rewards for high-frequency users, and merchant-specific deals for category-heavy spenders. Segment drift monitoring triggers re-segmentation when user behavior shifts significantly.

Limitation: K-Means requires a predefined number of clusters *k*, which may not reflect true underlying structure. Poor initialization or an incorrect *k* can produce misleading segments, requiring careful validation using silhouette scores and domain expert review.

Problem 2: Fraud and Anomaly Detection

Problem Definition: Fraudulent transactions — account takeovers, unusual transfer spikes, or device spoofing — are rare but costly. Since fraud patterns constantly evolve, supervised models require labeled data that quickly becomes stale. The goal is to detect anomalous transactions in near real-time without relying on historical fraud labels.

Proposed Approach: Autoencoders (deep learning-based dimensionality reduction) are trained exclusively on legitimate transaction data. Fraudulent transactions produce high reconstruction errors, flagging them as anomalies. For lightweight real-time deployment, Isolation Forest complements Autoencoders by efficiently isolating outliers in high-dimensional feature spaces.

Business Integration: Models are deployed in a real-time streaming pipeline (e.g., Apache Kafka + model serving API). Suspicious transactions are automatically held for review or trigger step-up authentication (OTP re-verification), while a risk score is logged for analyst dashboards.

Limitation: Autoencoders may generate false positives during legitimate unusual events (festival bulk payments, salary disbursements), requiring dynamic threshold tuning and contextual business rules to avoid unnecessary friction for genuine users.

2. Short Questions (Answer 1 from each category — 5 Marks total)

8

Category 1: Overfitting (2.5 Marks)

Option 2: Techniques to Reduce Overfitting

Overfitting is a common challenge in deep learning, especially when models have high capacity.

• Describe at least two techniques used to reduce overfitting (e.g., dropout, early stopping, data augmentation, L2 regularization).

• For each technique:

– Explain the intuition behind how it works.

– Describe how it affects the training process or model behavior.

– Provide a real-world application example (e.g., image classification, text classification, fraud detection).

⇨

11 Two powerful techniques to prevent overfitting are Dropout and Early Stopping.

Dropout is a regularization method used during neural network training, which involves dropping out certain nodes. It randomly turns off a portion of the neurons in a layer during each step of training, thus preventing the network from becoming too dependent on any particular neuron. The idea is that it is similar to training a large number of different "thinned" networks. Later, when the network predicts, all the neurons are active, but they are scaled down and thus represent an average of many models. A neuron, for instance, in an image classifier, might memorize a background feature, such as "blue sky", if each of the training images of birds contains a blue sky. When you remove the dependency, the network has to learn the shape of the bird, and does a much better job of that on new images. This is a very simple method, which has the effect of dramatically lowering co-adaptation of neurons.

Overfitting is addressed in a different way in Early Stopping. It will always be decreasing on the training set, but as soon as the model starts to memorize noise, the error on the held-out validation set will start to increase. Early Stopping just keeps track of the validation loss and stops training when the validation loss does not get better and starts to get worse. An actual application is to predict the stock price using a deep model. If the model is not stopped early, it could run too many iterations and pick up only the random daily jitter from the training period. Early stopping ensures that we retain a model that has learned the general trend in the data and will be useful for future data that was not included in the training set.

4

Category 2: Neural Network Architecture (2.5 Marks)

Option 1: CNN vs RNN

Explain the fundamental differences between a Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN):

• In terms of:

– Input structure and data type

– Information flow and architecture design

– Parameter sharing mechanism

• Provide suitable real-world examples where each architecture is preferred (e.g., image classification for CNNs, sequential/time-series modeling for RNNs).

• Briefly discuss common challenges encountered during training deep learning models, such as:

– Vanishing or exploding gradients

– Overfitting

• For each challenge, mention at least one mitigation technique (e.g., batch normalization, gradient clipping, early stopping).

⇨

CNNs and RNNs, they are pretty different in how they are set up for handling data. I mean, a CNN is mostly for stuff like images, where things are arranged in a grid. The way it works involves this convolution thing, basically a filter moving over the input to pick out local stuff like edges. It builds up from simple patterns to bigger ones in a straight feedforward path. What makes it efficient is sharing the same filter across the whole image, so it spots patterns no matter where they are. Like, ResNet is a good example for classifying images.

That spatial sharing helps a lot. RNNs are the opposite, they deal with sequences, text or time series data. The key part is this hidden state that carries over from one step to the next, kind of remembering what came before. So information flows through time, each output depending on the now and the past buildup in that state. Parameters get shared over time too, same weights every step, which lets it handle sequences that are long or short without much trouble.

But simple RNNs have this vanishing gradient issue, where early parts of the sequence lose their signal during training, and it fades out fast. That makes it hard to learn connections over long distances. I think CNNs do not run into that as much with their depth, though they can get heavy on computation from all those sliding filters. It seems like each has its own trade offs, depending on the data. CNNs build hierarchies spatially, while RNNs loop temporally. Some people might say one is better for certain tasks, but it gets a bit messy figuring out hybrids or something. Anyway, the position invariance in CNNs stands out, that is easy to miss when comparing.